Computer Networks
Fall 2024/25

Exercise-4

Submission in pairs by Thursday, **16-1-2025.** Submission is done by uploading your work to the course Moodle website. >>> No late submissions will be accepted
The name of the submitted file must be Exercise4_username1_username2**.pdf**
For example: Exercise4_israel.israeli_david.davidi.pdf (with dots). The first and last names of both students must appear. Only one student should submit the assignment (not both). It Covers Chapter 4 from the textbook. Please notice that once the grades for an exercise are published, appeals can only be submitted in the following five days. The grader is Mr. Daniel Ohayon, and you may contact him by email at *reichman.computer.networks@gmail.com.*

Problem 1 (20 points).
Consider sending a 3480-byte datagram (which includes the IP header) into a link that has an MTU of 700 bytes (700 bytes includes the IP header). Suppose the original datagram is stamped with the identifier number 1900. How many fragments are generated? What are their characteristics (i.e., Identification numbers, sizes, offsets, flags)? The table's number of rows might differ from the number of fragments.
Briefly justify your answer.

Answer:
The number of required fragments is _____

Fragments (packet) number	**Identification number**	**Flag**	**Offset**	**Size of the fragment (Including IP headers)**
1				

Problem 2 (20 points).
Consider the following routing table:

Address prefix	Next hop
196.94.0.0/16	A
196.94.2.0/24	B
196.94.2.128/26	C
196.94.2.192/26	D
196.94.64.0/18	E
196.76.0.0/14	F
140.0.0.0/8	G
128.0.0.0/2	H
96.0.0.0/2	I
0.0.0.0/0	J

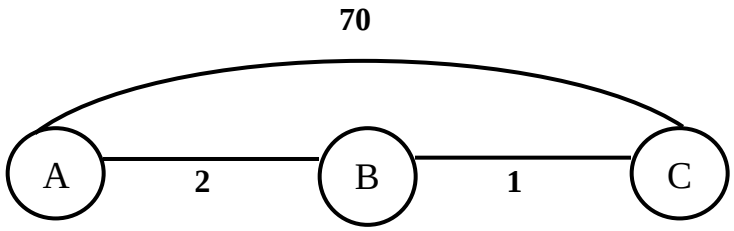
State the next hop for the following destination addresses.

Question #	Destination IP address	Answer (one letter)
1	32.2.3.4	
2	196.94.2.116	
3	196.94.2.129	
4	139.1.1.1	
5	196.94.3.115	
6	64.111.2.0	

Justify your answer. (The answer should be at most 10 lines long.)

Problem 3 (20 points).
A router continuously sends IP packets whose total length (data plus headers) is 512 **bytes**.
Assuming that packets live 100 seconds, what is the maximum transmission rate
(in **bits**/sec) that the router can operate without the danger of cycling through the IP datagram 16-bit identifier number space? Briefly justify your answer.
The answer should be at most 6 lines long.
Hint: The maximum transmission rate is more than 1,000,000 bps.

Problem 4 (20 points).
The question refers to the figure below when routing is done using distance vector.



1. What is the distance vector table of each of the three nodes when the network stabilizes **without** using poisoned reverse (i.e., when no more messages are sent)?
2. What is the distance vector table of each of the three nodes when the network stabilizes **with** poisoned reverse (i.e., when no more messages are sent)?

Assumption. Let's assume that after the network stabilizes (i.e., no more messages are sent), the weight on the edge AB changes **from 2 to 150**.
Definition. A *round* means that all the nodes that have messages to send, send them to their neighbors, get updates, and re-compute their distance vector tables.

3. How many extra rounds of sending messages are necessary before the network stabilizes again, assuming routing is done using distance vector **without** poisoned reverse?
4. How many extra rounds of sending messages are necessary before the networks stabilize again, assuming routing is done using distance vector **with** poisoned reverse?
Justify your answers.

Problem 5 (20 points).
A large number of consecutive IP addresses are available, starting at 108.24.208.0. Suppose eight organizations, A,B,C,D E,F,G and H, request 512, 4096, 16384, 4096, 2048, 1024, 1024, and 512 addresses, respectively, and in that order. For organizations A,B,C,D,E,F,G,H what is the first IP address assigned, the last IP address assigned, and the mask in the *w.x.y.z/s* notation? Write the addresses in the table below.
Important: at any time, assign the smallest available IP addresses!

Organization	First address	Last address	How many	Mask
A			**512**	
B			**4096**	
C			**16384**	
D			**4096**	
E			**2048**	
F			**1024**	
G			**1024**	
H			**512**	